

Galactic Binary Residuals in LISA SGWB Amplitude Inference Source Power Concentration and Spectral Degeneracy

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Outline

- 1 The Problem
- 2 Data & Method
- 3 Results
- 4 Conclusions

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- 1 The Problem
 - Motivation
 - Scientific question
- 2 Data & Method
- 3 Results
- 4 Conclusions

The Galactic confusion foreground

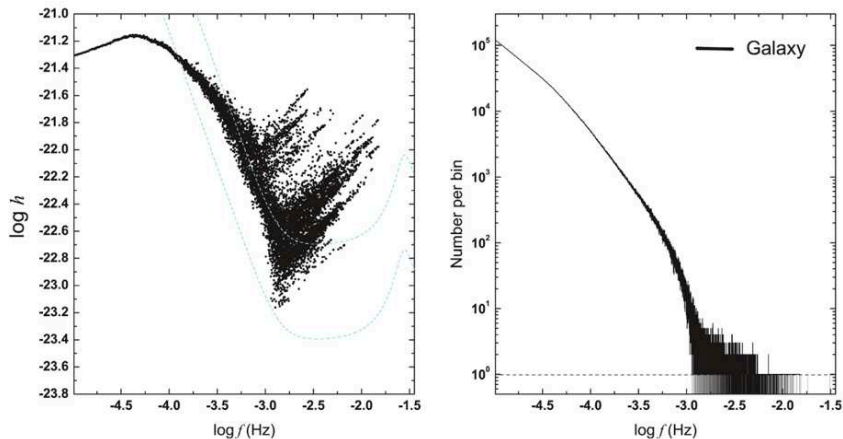


Figure: Yu & Jeffery, A&A (2010).

One residual, two statistical questions

Q1: Source discreteness

Fourth-order source statistic

Within one frequency bin

Many comparable sources



a few dominant sources

How concentrated is the residual
power?

Q2: Spectral degeneracy

Mean binned power

Across the frequency band

Residual spectrum



SGWB spectrum

How does residual uncertainty
affect the inferred SGWB
amplitude?

From the Galactic foreground to a catalog residual

LDC2A Sangria injections



Erebor comparison table



Catalog residual

- Sources not recovered:
 - Recovery confidence $C_{\text{rec}} < 0.9$
 - Injection–recovery overlap $\mathcal{O}_{\text{best}} < 0.9$

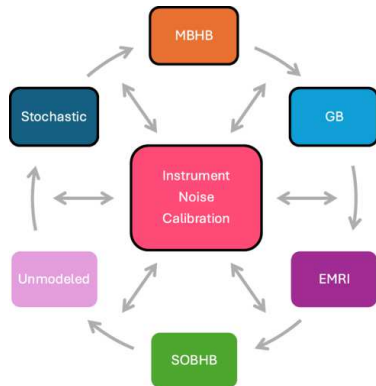
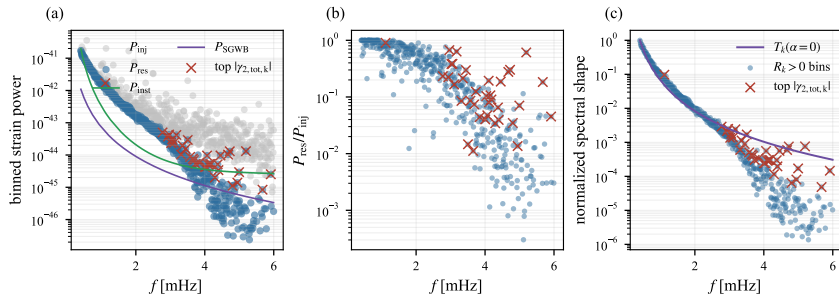


Figure: Katz et al., PRD (2025).

The residual and SGWB occupy the same frequencies



$$\Omega_{\text{GW}}(f) = \Omega_0 \left(\frac{f}{f_{\text{ref}}} \right)^\alpha, \quad f_{\text{ref}} = 3 \text{ mHz}, \quad S_h \propto \Omega_{\text{GW}}/f^3.$$

Fiducial: $\Omega_0 = 10^{-11}$, $\alpha = 0$

R_k and $T_k(0)$ share a low-frequency shape.

Outline

1 The Problem

2 Data & Method

- Catalog residual
- Q1: Source discreteness
- Q2: Spectral degeneracy

3 Results

4 Conclusions

From catalog sources to binned power

- Analysis band: 0.4–6.0 mHz
- $\Delta f = 10^{-5}$ Hz $\Rightarrow$ 560 bins
- 487 bins with $R_k > 0$

Catalog residual template

$$R_k = \sum_{i \in k, \text{res}} p_i$$

SGWB template

$$T_k(\alpha) = \sum_{j \in k} \frac{3H_0^2}{2\pi^2 f_j^3} \left(\frac{f_j}{f_{\text{ref}}} \right)^\alpha \delta f_F$$

Mean binned power model

$$\mu_k = P_{\text{inst},k} + \Omega_0 T_k(\alpha) + \beta R_k$$

Ω_0 : SGWB amplitude (target).

β : residual-power factor (nuisance).

Ω_0 – β degeneracy from the spectral overlap of T_k and R_k .

Prior width σ_β : $\beta = 1 \pm \sigma_\beta$.

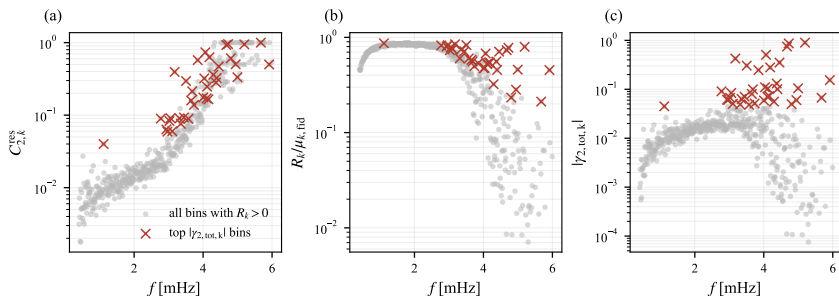
How well must the residual power be known?

Source concentration and non-Gaussianity

Excess kurtosis γ_2 : measures deviation from Gaussianity.

$\gamma_2 = 0$ for Gaussian; $\gamma_2 < 0$ means flatter-than-Gaussian tails.

$$C_{2,k} = \sum_i w_i^2 \quad \Rightarrow \quad \gamma_{2,\text{res},k} = -\frac{3}{2}C_{2,k} \quad \Rightarrow \quad \gamma_{2,\text{tot},k} = \gamma_{2,\text{res},k} \left(\frac{R_k}{\mu_{k,\text{fid}}} \right)^2$$



Median $|\gamma_{2,\text{tot}}| \simeq 1.4 \times 10^{-2}$.

Quantifying the degeneracy: Fisher and marginalization

Spectral overlap $\rho_{\Omega\beta}$

Fisher elements:

$$A = \sum \frac{T_k^2}{\text{Var}_k}, \quad B = \sum \frac{R_k^2}{\text{Var}_k}$$

$$C = \sum \frac{T_k R_k}{\text{Var}_k}$$

$$\rho_{\Omega\beta} = \frac{C}{\sqrt{AB}}$$

No-prior uncertainty increase

$$I_0 = \frac{\sigma_{\Omega, \text{marg}}}{\sigma_{\Omega, \text{fixed}}} = \frac{1}{\sqrt{1 - \rho_{\Omega\beta}^2}}$$

- $\rho = 0 \Rightarrow I_0 = 1$: orthogonal.
- $|\rho| \rightarrow 1 \Rightarrow I_0 \rightarrow \infty$: degenerate.

Prior width condition: $\beta = 1 \pm \sigma_\beta \Rightarrow B \rightarrow B + 1/\sigma_\beta^2$.

Choose σ_β so that $I(\sigma_\beta) \leq r_{\text{max}}$.

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2 Data & Method

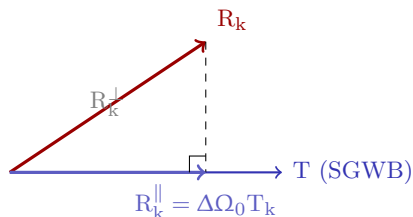
3 Results

- All-bin result
- Frequency structure

4 Conclusions

Omitting vs. marginalizing the residual

Wrong model: omit R_k



$$\Delta\Omega_0 = \frac{C}{A} \simeq 2.5 \times 10^{-11}$$

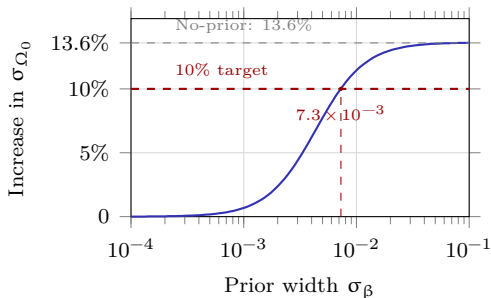
Residual absorbed as extra SGWB power.

$$\Omega_{0,\text{fit}} \simeq 3.5 \times 10^{-11}$$

$$(\Omega_{0,\text{true}} = 10^{-11}).$$

119.5 $\times$ the fixed- β Fisher uncertainty.

Correct model: marginalize β



$\sigma_\beta \rightarrow 0 \Rightarrow I \rightarrow 1$ (fixed β).
 $\sigma_\beta \rightarrow \infty \Rightarrow I \rightarrow I_0 = 1.136$ (no prior).

Robustness to unresolved sub-bin frequency structure

Catalog-to-Fourier mapping ($\Delta f = 10^{-5}$ Hz, $\delta f_F \simeq 3.2 \times 10^{-8}$ Hz, 315–316 modes/bin). Sets both R_k (mean) and mode-power variance (covariance).

Uniform within bin (coarse-grained reference)



schematic

Evenly distributed across ~ 315 modes.

Drift + Doppler (physically motivated)



schematic

Within chirp + annual Doppler range.

Mapping	$\min_k \nu_{\text{eff},k}$	Inflation	$\sigma_{\beta, \text{max}}^{10\%}$
Uniform within bin	~ 630	13.6%	0.0073
Drift + Doppler support	~ 93	11.2%	0.0145

ν_{eff} : moment-matched chi-square d.o.f. Closest-frequency stress test in backup.

Finite-source random-phase diagonal: 13.6% $\rightarrow$ 13.8%. Reference result not driven by Gaussian self-noise (diagonal only; cross-frequency covariance not included).

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 - Takeaways
 - Scope

Summary

- 1 Residual omission causes parameter bias.
The fitted Ω_0 absorbs residual power aligned with the SGWB template. $\Delta\Omega_0/\Omega_{0,\text{true}} \simeq 2.5$.
- 2 Residual uncertainty propagates into SGWB uncertainty.
Marginalizing β opens a spectral-degeneracy direction: 13.6% no-prior inflation, $\sigma_{\beta,\text{max}}^{10\%} \simeq 0.0073$.
- 3 Gaussian residual self-noise does not drive the reference result.
Finite-source random-phase diagonal covariance: 13.6% $\rightarrow$ 13.8%.

Reference case: $\alpha = 0$, $\Omega_0 = 10^{-11}$, all 560 bins, uniform within each bin.

Scope and future work

Current model

- Catalog residual from recovery-threshold classification
- Orbit-averaged scalar Michelson X power
- Independent complex Gaussian Fourier amplitudes
- Diagonal covariance fixed at the fiducial spectrum
- One residual normalization parameter β

Future work

- Global-fit residual realizations and subtraction uncertainty
- Time-dependent, multichannel TDI response
- Coherent cross-frequency and cross-channel covariance
- Flexible residual spectral and population-level model

Interpretation boundary

The reported numbers are conditional on a catalog-level scalar-power model; they are not a full coherent, multichannel LISA posterior forecast.

Thank You

Comments and questions are warmly welcome.

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Backup Slides

All-bin Fisher details

Gaussian residual amplitudes

Residual distribution	$ \rho_{\Omega\beta} $	I_0	$\sigma_{\beta}^{10\%}$	Min. d.o.f.
Uniform within each bin	0.475	1.136	0.00728	~ 630
Intrinsic drift + Doppler	0.438	1.112	0.01451	~ 93
Closest Fourier frequency	0.308	1.051	No prior	~ 3.3

- Uniform finite-source diagonal check: $I_0 = 1.138$ and $\sigma_{\beta}^{10\%} = 0.00698$.
- Reference case: $\Omega_{0,\text{fid}}/\sigma_{\Omega,\text{fixed}} = 47.9$.
- The 73 bins with $R_k = 0$ add $F_{\Omega\Omega}$ but not $F_{\Omega\beta}$ or $F_{\beta\beta}$.
- Reference omitted-residual shift: $\Delta\Omega_{\text{bias}}/\sigma_{\Omega,\text{fixed}} = 119.5$.

Overall noise-amplitude uncertainty

$$\mu_k = \eta P_{\text{inst},k} + \Omega_0 T_k + \beta R_k, \quad \eta_{\text{fid}} = 1$$

Quantity	Fixed η	η marginalized
$\Omega_{0,\text{fid}}/\sigma_{\Omega,\text{fixed}}$	47.9	17.0
I_0 from marginalizing β	1.136	1.130
$\sigma_{\beta}^{10\%}$	0.00728	0.00848

Uncertainty in the overall instrumental-noise amplitude affects the absolute Ω_0 constraint more than the relative cost of marginalizing β .

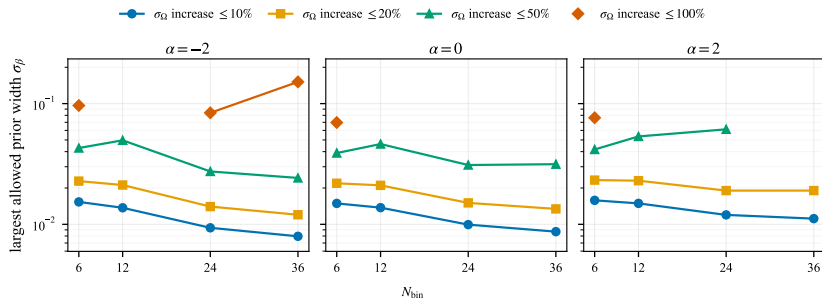
Selected frequency subsets

487 bins with $R_k > 0$, ranked by $|\gamma_{2,\text{tot},k}|$:

α	N_{bin}	$\Omega_{0,\text{fid}}/\sigma_{\Omega,\text{fixed}}$	I_0	$\sigma_{\beta}^{10\%}$
-2	6	1.28	2.36	0.015
-2	36	6.98	2.03	0.008
0	6	2.29	3.04	0.015
0	12	4.73	1.84	0.014
0	24	7.62	1.97	0.010
0	36	9.98	1.76	0.009
2	6	4.52	2.90	0.016
2	36	17.51	1.42	0.011

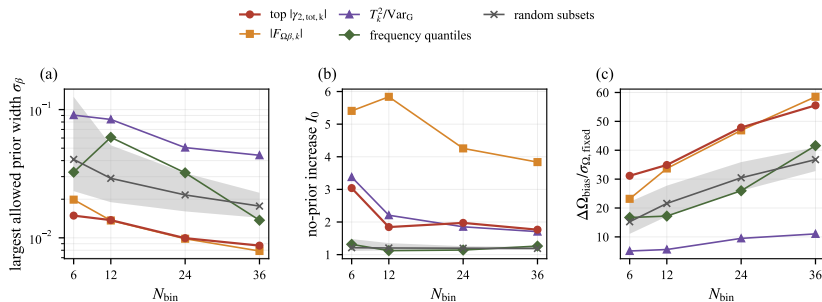
The ranking selects bins where the random-phase source discreteness remains least diluted; it does not rank the spectral overlap itself.

Prior widths for selected subsets



$\alpha = -2, 0, 2$; thresholds of 10%, 20%, 50%, 100%. Open triangles indicate that no informative prior is required.

Dependence on frequency-bin selection



Gray bands: 16th–84th percentiles from 200 random subsets at each N_{bin} ; they show variation across frequency subsets, not measurement uncertainty.

Distinct but connected statistical objects

$\gamma_{2,\text{tot}}$ describes the distribution of an auxiliary, response-averaged random-phase source sum. It does not determine the full cross-frequency covariance of Fourier power.

- C_2 and $\gamma_{2,\text{tot}}$ are used to characterize source discreteness.
- For the uniform-within-bin mapping, $C_{2,k}$ determines the finite-source diagonal correction to the mode-power covariance.
- The all-bin Fisher result uses a fixed Gaussian mode-power covariance; the finite-source check changes 13.6% to 13.8%.
- Subsets are ranked by $|\gamma_{2,\text{tot},k}|$ only to select bins where the random-phase signature remains least diluted.

Random phase model derivation

Model: $x_{\text{res}} = \sum_i x_i$, $x_i = \sqrt{2p_i} \cos \phi_i$, $\phi_i \sim U(0, 2\pi)$ i.i.d.

Moments of a single term: $\langle \cos \phi \rangle = 0$, $\langle \cos^2 \phi \rangle = \frac{1}{2}$, $\langle \cos^4 \phi \rangle = \frac{3}{8}$.

$$\langle x_i \rangle = 0, \quad \langle x_i^2 \rangle = p_i, \quad \langle x_i^4 \rangle = \frac{3}{2} p_i^2$$

$$\kappa_{4,i} = \langle x_i^4 \rangle - 3 \langle x_i^2 \rangle^2 = -\frac{3}{2} p_i^2$$

Cumulants add for independent variables: $\kappa_{4,\text{res}} = -\frac{3}{2} \sum_i p_i^2$, $\sigma_{\text{res}}^2 = \sum_i p_i$.

$$\text{Excess kurtosis: } \gamma_{2,\text{res}} = \frac{\kappa_{4,\text{res}}}{\sigma_{\text{res}}^4} = -\frac{3}{2} \frac{\sum p_i^2}{(\sum p_j)^2} = \boxed{-\frac{3}{2} C_2}$$

Note: p_i already contains orbital average $\rightarrow x_{\text{res}}$ is auxiliary response-averaged source sum, not time-dependent Michelson strain.

Noise dilution step-by-step

Setup: $\mathbf{x} = \mathbf{x}_{\text{res}} + \mathbf{x}_{\text{G}}$ ($\mathbf{x}_{\text{G}}$ approx. Gaussian). Cumulants add for independent variables.

Gaussian properties: $\kappa_{4,\text{G}} = 0$.

$$\begin{aligned}\kappa_{4,\text{tot}} &= \kappa_{4,\text{res}} = \gamma_{2,\text{res}} \sigma_{\text{res}}^4 \\ \sigma_{\text{tot}}^2 &= \sigma_{\text{res}}^2 + \sigma_{\text{G}}^2\end{aligned}$$

Dilution formula:

$$\gamma_{2,\text{tot}} = \frac{\kappa_{4,\text{tot}}}{\sigma_{\text{tot}}^4} = \gamma_{2,\text{res}} \frac{\sigma_{\text{res}}^4}{(\sigma_{\text{res}}^2 + \sigma_{\text{G}}^2)^2} = \gamma_{2,\text{res}} \left(\frac{\sigma_{\text{res}}^2}{\sigma_{\text{tot}}^2} \right)^2$$

Identifying $\sigma_{\text{res}}^2 = R_{\mathbf{k}}$, $\sigma_{\text{tot}}^2 = \mu_{\mathbf{k},\text{fid}}$: $\gamma_{2,\text{tot},\mathbf{k}} = \gamma_{2,\text{res},\mathbf{k}} (R_{\mathbf{k}}/\mu_{\mathbf{k},\text{fid}})^2$.

Fourier-mode covariance construction

Sangria parameters: $N = 6\,307\,200$, cadence 5 s , $T_{\text{obs}} = 31\,536\,000\text{ s}$,
 $f_j = j/T_{\text{obs}}$, $\delta f_F = 3.17 \times 10^{-8}\text{ Hz}$.

Independent Gaussian Fourier-mode powers: $\langle \hat{S}_j \rangle = S_j$, $\text{Cov}(\hat{S}_j, \hat{S}_{j'}) = S_j^2 \delta_{jj'}$.

Binned power: $\hat{V}_k = \delta f_F \sum_{j \in \mathcal{J}_k} \hat{S}_j$, $\mu_k = \delta f_F \sum_j S_j$, $\text{Var}_G = (\delta f_F)^2 \sum_j S_j^2$.

PSD decomposition: $S_j = S_{\text{inst}}(f_j) + \Omega_0 \mathcal{T}_j(\alpha) + \beta S_{\text{res},j}$.

Three residual Fourier distributions (same total catalog power):

- Uniform: $\nu_{\text{eff}} \approx 2M_k \approx 630$ (CLT well-satisfied)
- Drift+Doppler (chirp+Doppler broadening): $\nu_{\text{eff}} \approx 93$
- Closest-frequency: $\nu_{\text{eff}} \approx 3.3$ (Gaussian approximation weakest; limiting case)

The assigned powers determine both R_k in the mean and the covariance; drift–Doppler intervals can transfer power between neighboring bins.

Residual covariance model dependence

Let G_j be the instrumental+SGWB spectral density and H_j the residual spectral density at Fourier frequency j .

Gaussian residual amplitudes

$$\text{Var}(\hat{S}_j) = (G_j + H_j)^2$$

Includes the H_j^2 variance from Gaussian residual amplitudes.

Finite-source random phase

$$\text{Var}(\hat{S}_j) = (G_j + H_j)^2 - C_{2,k} H_j^2$$

Uniform distribution; diagonal correction.

Fixed residual Fourier realization

$$\text{Var}(\hat{S}_j) = G_j^2 + 2G_j H_j$$

Conditional on a fixed residual Fourier realization.

Residual distribution	Gaussian	Finite source	Fixed residual
Uniform within each bin	13.6%	13.8%	21.0%
Intrinsic drift + Doppler range	11.2%	—	21.1%
Closest Fourier frequency	5.1%	—	21.0%

Interpretation

The uniform all-bin increase changes by only 0.2 percentage points under the finite-source diagonal correction.

Prior widths for the uniform-within-bin mapping: 0.00728 (Gaussian), 0.00698 (finite source), and 0.00258 (fixed residual). None includes coherent cross-frequency covariance.

Instrumental-noise convention

Long-wavelength single-Michelson noise:

$$S_{\text{inst}}(f) = \frac{20}{3L^2} \left[P_{\text{OMS}}(f) + \frac{4P_{\text{acc}}(f)}{(2\pi f)^4} \right], \quad L = 2.5 \times 10^9 \text{ m}$$

OMS (position noise):

$$P_{\text{OMS}}(f) = (1.5 \times 10^{-11} \text{ m})^2 \text{ Hz}^{-1} [1 + (2.0 \times 10^{-3} \text{ Hz/f})^4]$$

Acceleration noise: $P_{\text{acc}}(f) =$

$$(3.0 \times 10^{-15} \text{ m s}^{-2})^2 \text{ Hz}^{-1} [1 + (0.4 \times 10^{-3} \text{ Hz/f})^2] [1 + (f/8.0 \times 10^{-3} \text{ Hz})^4]$$

Convention: $20/3 = 1/\mathcal{R}_{\text{LW}}^{\text{X}}$ converts to equivalent incident strain, same as p_i and SGWB template. $P_{\text{inst},k} = \delta f_{\text{F}} \sum_{j \in \mathcal{J}_k} S_{\text{inst}}(f_j)$.

Fisher matrix construction

Setup: $\theta = (\Omega_0, \beta)$, covariance diagonal and fixed at fiducial.

$$F_{ij} = \sum_k \frac{(\partial_{\theta_i} \mu_k)(\partial_{\theta_j} \mu_k)}{\text{Var}_G(\hat{V}_k)}, \quad \partial_{\Omega_0} \mu_k = T_k(\alpha), \quad \partial_{\beta} \mu_k = R_k$$

Element	Expression
$A = F_{\Omega\Omega}$	$\sum_k T_k^2 / \text{Var}_G(\hat{V}_k)$
$B = F_{\beta\beta}$	$\sum_k R_k^2 / \text{Var}_G(\hat{V}_k)$
$C = F_{\Omega\beta}$	$\sum_k T_k R_k / \text{Var}_G(\hat{V}_k)$

Gaussian prior $\beta = 1 \pm \sigma_{\beta}$: $B \rightarrow B_{\sigma} = B + 1/\sigma_{\beta}^2$. Fixed β :
 $\sigma_{\Omega, \text{fixed}} = A^{-1/2}$. Marginalized: $\sigma_{\Omega, \text{marg}} = \sqrt{B_{\sigma} / (AB_{\sigma} - C^2)}$.

Uncertainty increase and prior-width condition

Uncertainty inflation: $I^2(\sigma_\beta) = AB_\sigma / (AB_\sigma - C^2)$, $B_\sigma = B + 1/\sigma_\beta^2$.

No-prior limit: $I_0 = 1/\sqrt{1 - \rho^2}$ with $\rho = C/\sqrt{AB}$.

Prior-width condition $I(\sigma_\beta) \leq r_{\max}$:

$$B + \frac{1}{\sigma_\beta^2} \geq \frac{C^2}{A(1 - r_{\max}^{-2})}$$

- If $B \geq C^2/[A(1 - r_{\max}^{-2})]$: no informative prior needed
- Otherwise: $\sigma_{\beta, \max} = [C^2/(A(1 - r_{\max}^{-2})) - B]^{-1/2}$

Thresholds: $r_{\max} = 1.1, 1.2, 1.5, 2.0$ (10%, 20%, 50%, 100%).

Omitted-residual bias: $\Delta\Omega_{\text{bias}}/\sigma_{\Omega, \text{fixed}} = \beta_{\text{fid}} C/\sqrt{A}$. (Linearized shift — distinct from marginalization penalty.)

Random-phase moment comparison

Procedure: 5×10^4 realizations per bin, 487 bins with $R_k > 0$. For each, draw $\phi_i \sim U(0, 2\pi)$ i.i.d., compute $x_{\text{res}} = \sum_i \sqrt{2p_i} \cos \phi_i$, estimate γ_2 from sample moments.

Metric	Value
Pearson $r(\gamma_{2,\text{MC}}, \gamma_{2,\text{res}})$	0.9991
RMS($\Delta\gamma_2$)	1.78×10^{-2}
95th percentile $ \Delta\gamma_2 $	3.71×10^{-2}
Range of $\gamma_{2,\text{res}}$	$[-1.5, 0]$

Wider scatter at small C_2 : sampling variance near Gaussian limit.
 $\text{corr}(\Delta\gamma_2, C_2) = 0.036$ — no systematic bias.

Analysis settings reference

Setting	Value
Catalog data	LDC2A Sangria + Erebor comparison table
Response	Orbit-averaged Michelson X, $\mathcal{R}_{\text{LW}}^{\text{X}} = 3/20$
Frequency range	0.4–6.0 mHz
Bin width Δf	10^{-5} Hz (560 bins)
Observation	$T_{\text{obs}} = 31\,536\,000$ s, cadence 5 s
Fourier modes per bin	315 or 316
Recovery criteria	$C_{\text{rec}} \geq 0.9$, $\mathcal{O}_{\text{best}} \geq 0.9$
Fiducial SGWB	$\Omega_0 = 10^{-11}$, $\alpha = -2, 0, 2$, $f_{\text{ref}} = 3$ mHz
H_0	$67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$
Noise model	Long-wavelength single-Michelson form
Covariance	Gaussian-amplitude; finite-source and fixed-residual comparison
Fisher parameters	(Ω_0, β) , Gaussian prior on β
Thresholds r_{max}	1.1, 1.2, 1.5, 2.0